

Задачи по линейной алгебре с SEEMOUS

1. (2007) Let $f(\mathbf{x}) = \max_{1 \leq i \leq n} |x_i|$ for $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ and let A be an $n \times n$ matrix such that $f(A\mathbf{x}) = f(\mathbf{x})$ for all $\mathbf{x} \in \mathbb{R}^n$. Prove that there exists a positive integer m such that A^m is the identity matrix I_n .
2. (2008) Let $M_n(\mathbb{R})$ denote the set of all real $n \times n$ matrices. Find all surjective functions $f : M_n(\mathbb{R}) \rightarrow \{0, 1, \dots, n\}$ which satisfy

$$f(XY) \leq \min\{f(X), f(Y)\}$$

for all $X, Y \in M_n(\mathbb{R})$.

3. (2009) Let $\text{SL}_2(\mathbb{Z}) = \{A \mid A \text{ is a } 2 \times 2 \text{ matrix with integer entries and } \det A = 1\}$.
 - a) Find an example of matrices $A, B, C \in \text{SL}_2(\mathbb{Z})$ such that $A^2 + B^2 = C^2$.
 - b) Show that there do not exist matrices $A, B, C \in \text{SL}_2(\mathbb{Z})$ such that $A^4 + B^4 = C^4$.
4. Given the real numbers $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ we define the $n \times n$ matrices $A = (a_{ij})$ and $B = (b_{ij})$ by

$$a_{ij} = a_i - b_j \quad \text{and} \quad b_{ij} = \begin{cases} 1, & \text{if } a_{ij} \geq 0, \\ 0, & \text{if } a_{ij} < 0, \end{cases}$$

for all $i, j \in \{1, 2, \dots, n\}$.

Consider $C = (c_{ij})$ a matrix of the same order with elements 0 and 1 such that

$$\sum_{j=1}^n b_{ij} = \sum_{j=1}^n c_{ij}, \quad i \in \{1, 2, \dots, n\} \quad \text{and} \quad \sum_{i=1}^n b_{ij} = \sum_{i=1}^n c_{ij}, \quad j \in \{1, 2, \dots, n\}.$$

Show that:

a)

$$\sum_{i,j=1}^n a_{ij}(b_{ij} - c_{ij}) = 0 \quad \text{and} \quad B = C.$$

b) B is invertible if and only if there exist two permutations σ and τ of the set $\{1, 2, \dots, n\}$ such that

$$b_{\tau(1)} \leq a_{\sigma(1)} < b_{\tau(2)} \leq a_{\sigma(2)} < \dots \leq a_{\sigma(n-1)} < b_{\tau(n)} \leq a_{\sigma(n)}.$$

5. (2010) Denote by $\mathcal{M}_2(\mathbb{R})$ the set of all 2×2 matrices with real entries. Prove that:
 - a) for every $A \in \mathcal{M}_2(\mathbb{R})$ there exist $B, C \in \mathcal{M}_2(\mathbb{R})$ such that $A = B^2 + C^2$;
 - b) there do not exist $B, C \in \mathcal{M}_2(\mathbb{R})$ such that

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = B^2 + C^2$$

and $BC = CB$.

6. Suppose that A and B are $n \times n$ matrices with integer entries, and let $B \neq 0$. Prove that there exists $m \in \mathbb{N}$ such that the product AB^{-1} can be represented as

$$AB^{-1} = \sum_{k=1}^m N_k^{-1},$$

where N_k are $n \times n$ matrices with integer entries for all $k = 1, \dots, m$, and $N_i \neq N_j$ for $i \neq j$.

7. (2011) Let $A = (a_{ij})$ be a real $n \times n$ matrix such that $A^n \neq 0$ and $a_{ij}a_{ji} \leq 0$ for all i, j . Prove that there exist two nonreal numbers among eigenvalues of A .
8. (2012) Let $A = (a_{ij})$ be the $n \times n$ matrix, where a_{ij} is the remainder of the division of $i^j + j^i$ by 3, for $i, j = 1, 2, \dots, n$. Find the greatest n for which $\det A \neq 0$.
9. a) Prove that if k is an even positive integer and A is a real symmetric $n \times n$ matrix such that

$$(\operatorname{Tr}(A^k))^{k+1} = (\operatorname{Tr}(A^{k+1}))^k,$$

then

$$A^n = \operatorname{Tr}(A) A^{n-1}.$$

- b) Does the assertion from part (a) also hold for odd positive integers k ?

10. (2013) Let $M, N \in M_2(\mathbb{C})$ be two nonzero matrices such that

$$M^2 = N^2 = 0_2 \quad \text{and} \quad MN + NM = I_2,$$

where 0_2 is the 2×2 zero matrix and I_2 the 2×2 unit matrix. Prove that there exists an invertible matrix $A \in M_2(\mathbb{C})$ such that

$$M = A \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} A^{-1} \quad \text{and} \quad N = A \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} A^{-1}.$$

11. Let $A \in M_n(\mathbb{Q})$ be such that there exists a positive integer m with $A^m = -I_n$. Prove that either $A^2 = -I_n$ or $A^3 = -I_n$.
12. (2014) Let n be a nonzero natural number and $f : \mathbb{R} \rightarrow \mathbb{R} \setminus \{0\}$ be a function such that $f(2014) = 1 - f(2013)$. Let $x_1, x_2, x_3, \dots, x_n$ be real numbers not equal to each other. If

$$\begin{vmatrix} 1 + f(x_1) & f(x_2) & f(x_3) & \cdots & f(x_n) \\ f(x_1) & 1 + f(x_2) & f(x_3) & \cdots & f(x_n) \\ f(x_1) & f(x_2) & 1 + f(x_3) & \cdots & f(x_n) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ f(x_1) & f(x_2) & f(x_3) & \cdots & 1 + f(x_n) \end{vmatrix} = 0,$$

prove that f is not continuous.

13. Let $A \in M_n(\mathbb{C})$ and $a \in \mathbb{C}$, $a \neq 0$ such that $A - A^* = 2aI_n$, where $A^* = (\bar{A})^t$ and $\bar{A}$ is the conjugate of the matrix A .

(a) Show that $|\det A| \geq |a|^n$.

(b) Show that if $|\det A| = |a|^n$ then $A = aI_n$.

14. (2015) For an integer $n > 2$, let $A, B, C, D \in M_n(\mathbb{R})$ be matrices satisfying:

$$AC - BD = I_n, \quad AD + BC = O_n,$$

where I_n is the identity matrix and O_n is the zero matrix in $M_n(\mathbb{R})$. Prove that:

a) $CA - DB = I_n$ and $DA + CB = O_n$,

b) $\det(AC) \geq 0$ and $(-1)^n \det(BD) \geq 0$.

15. (2016)

a) Prove that for every matrix $X \in M_2(\mathbb{C})$ there exists a matrix $Y \in M_2(\mathbb{C})$ such that $Y^2 = X^2$.

b) Prove that there exists a matrix $A \in M_3(\mathbb{C})$ such that $Z^2 \neq A^2$ for all $Z \in M_3(\mathbb{C})$.

16. Let $A_1, A_2, \dots, A_k$ be idempotent matrices (i.e., $A_i^2 = A_i$) in $M_n(\mathbb{R})$. Prove that

$$\sum_{i=1}^k N(A_i) \geq \text{rank} \left(I - \prod_{i=1}^k A_i \right),$$

where $N(A_i) = n - \text{rank}(A_i)$ and $M_n(\mathbb{R})$ is the set of square $n \times n$ matrices with real entries.

17. (2017) Let $A \in \mathcal{M}_2(\mathbb{R})$. Suppose

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

satisfies

$$a^2 + b^2 + c^2 + d^2 < \frac{1}{5}.$$

Show that $I + A$ is invertible.

18. Let $A, B \in \mathcal{M}_n(\mathbb{R})$.

a) Prove that there exists $a > 0$ such that for every $\varepsilon \in (-a, a)$, $\varepsilon \neq 0$, the matrix equation

$$AX + \varepsilon X = B, \quad X \in \mathcal{M}_n(\mathbb{R}),$$

has a unique solution $X(\varepsilon) \in \mathcal{M}_n(\mathbb{R})$.

b) Prove that if $B^2 = I_n$ and A is diagonalisable, then

$$\lim_{\varepsilon \rightarrow 0} \varepsilon \cdot \text{Tr}(BX(\varepsilon)) = n - \text{rank } A.$$

19. (2018) Let $m, n, p, q \geq 1$ and let the matrices $A \in M_{m,n}(\mathbb{R})$, $B \in M_{n,p}(\mathbb{R})$, $C \in M_{p,q}(\mathbb{R})$, $D \in M_{q,m}(\mathbb{R})$ be such that

$$A^t = BCD, \quad B^t = CDA, \quad C^t = DAB, \quad D^t = ABC.$$

Prove that $(ABCD)^2 = ABCD$.

20. Let $A, B \in M_{2018}(\mathbb{R})$ such that $AB = BA$ and $A^{2018} = B^{2018} = I$, where I is the identity matrix. Prove that if $\text{Tr}(AB) = 2018$, then $\text{Tr} A = \text{Tr} B$.
21. (2019) Let m, n be nonzero natural numbers. Show that for any matrices $A_1, \dots, A_m \in M_n(\mathbb{R})$ there exist $\varepsilon_1, \dots, \varepsilon_m \in \{-1, 1\}$ such that

$$\text{Tr}((\varepsilon_1 A_1 + \dots + \varepsilon_m A_m)^2) \geq \text{Tr}(A_1^2) + \dots + \text{Tr}(A_m^2). \quad (1)$$

22. Let $n \geq 2$ and $A, B \in M_n(\mathbb{C})$ with the property that $B^2 = B$. Show that

$$\text{rank}(AB - BA) \leq \text{rank}(AB + BA). \quad (1)$$

23. (2021) Let $n \geq 2$ be a positive integer and let $A \in \mathcal{M}_n(\mathbb{R})$ be a matrix such that $A^2 = -I_n$. If $B \in \mathcal{M}_n(\mathbb{R})$ and $AB = BA$, prove that $\det B \geq 0$.

24. Let $A \in \mathcal{M}_n(\mathbb{C})$ be a matrix such that $(AA^*)^2 = A^*A$, where $A^* = (\bar{A})^t$ denotes the Hermitian transpose (i.e., the conjugate transpose) of A .

(a) Prove that $AA^* = A^*A$.

(b) Show that the non-zero eigenvalues of A have modulus one.

25. (2022) Let $A, B \in M_n(\mathbb{C})$ be such that $AB^2A = AB$. Prove that:

a) $(AB)^2 = AB$.

b) $(AB - BA)^3 = O_n$.

26. Let $\alpha \in \mathbb{C} \setminus \{0\}$ and $A \in M_n(\mathbb{C})$, $A \neq O_n$, be such that

$$A^2 + (A^*)^2 = \alpha A \cdot A^*,$$

where $A^* = (\bar{A})^T$. Prove that $\alpha \in \mathbb{R}$, $|\alpha| \leq 2$, and $A \cdot A^* = A^* \cdot A$.

27. (2023) Prove that if A and B are $n \times n$ square matrices with complex entries satisfying

$$A = AB - BA + A^2B - 2ABA + BA^2 + A^2BA - ABA^2,$$

then $\det(A) = 0$.

28. Prove that: if A is an $n \times n$ square matrix with complex entries such that

$$A + A^* = A^2 A^*,$$

then $A = A^*$. (For any matrix M , denote by $M^* = \overline{M}^t$ the conjugate transpose of M .)

29. (2024) Let $A, B \in \mathcal{M}_n(\mathbb{R})$ be two real, symmetric matrices with nonnegative eigenvalues. Prove that

$$A^3 + B^3 = (A + B)^3 \quad \text{if and only if} \quad AB = \mathcal{O}_n.$$

30. Let $n \in \mathbb{N}$, $n \geq 2$. Find all the values $k \in \mathbb{N}$, $k \geq 1$, for which the following statement holds:

“If $A \in \mathcal{M}_n(\mathbb{C})$ is such that $A^k A^* = A$, then $A = A^*$.”

(Here, $A^* = \overline{A}^t$ denotes the conjugate transpose of A .)

31. (2025) Let A be an $n \times n$ matrix with strictly positive elements and two vectors $u, v \in \mathbb{R}^n$, also with strictly positive elements, such that

$$Au = v \quad \text{and} \quad Av = u.$$

Prove that $u = v$.

32. Let $A \in M_n(\mathbb{C})$ such that $A^* A^2 = A A^*$. Prove that $A^2 = A$. (We denote by $A^* = \overline{A}^t$ the conjugate transpose of A .)

33. (2026) Let A and B be $n \times n$ matrices with complex entries for which

$$A^2 + B^2 = 2(AB - BA)$$

Prove that $A^2 + B^2$ is a nilpotent matrix.

34. Let A and B be $n \times n$ matrices with complex entries for which $A^2 = AB - BA$ and $\text{rank}(A - B) = 1$. Prove that $ABA = \mathcal{O}_n$.